Week 2-4: Propositional Logic (Tutorial starts from Page 8)

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Contents

1 Syntax

Definition 1.1. *DSyntax decides what are allowed to be written*

1.1 Well-formed Formula (WFF)

A formula ψ is obtained by composing $p \in P$ with $c \in C$, where P is the set of propositions and C is the set of connectors. here we use any set of complete logic, like $C = \{\neg, \vee\}$. This is not a percise definition as counter examples that do not form a valid formula can be formulated based on this, for example $\neg \vee p$ means nothing. To formalize the definition we may attempt to define a language with the following grammar:

$$\Sigma = P \cup C \cup \{(,)\}$$

$$FORM : \{\psi = p \in P \mid \neg\psi \mid \psi \vee \psi \mid (\psi)\}$$

However this is a recursive definition as the definition of the a formula ψ involves itself. We need to find a declarative way to define this recursive/inductive construction.

Definition 1.2. *FORM is* **the smallest** *set of strings over* Σ *where*

1. $P \subseteq FORM$
2. $\forall\psi \in FORM, \neg\psi \in FORM$
3. $\forall\psi_1, \psi_2 \in FORM, \psi_1 \vee \psi_2 \in FORM$

We can see as a construction rule, we are admitting strings into the set of formula, but we need to somewhat impose restriction on what is not acceptable.

1.2 Closed Set

Let's remove the attributive **the smallest** first and define another set that contains all the set that satisfy these rules listed above:

Definition 1.3. *Closed is a set of strings over* Σ *where*

1. $P \subseteq Closed$
2. $\forall\psi \in Closed, \neg\psi \in Closed$
3. $\forall\psi_1, \psi_2 \in Closed, \psi_1 \vee \psi_2 \in Closed$

1.2.1 All Closed Set

And we define a set ACS (All-closed sets). We can define FORM via the following constriction:

$$FORM = \bigcap ACS$$

Suppose $BCS = \emptyset$ and from definition

$$U_0 = \bigcap BCS = \{w \in \Sigma^* \mid \forall S \in BCS, w \in S\}$$

However, $BCS = \emptyset$, The statement $\forall S \in BCS, w \in S$ is vacuously true $\forall w \in \Sigma^*$, which means that

$$U_0 = \Sigma^* = \bigcap BCS$$

A contradiction. Therefore we have proved that ACS cannot be empty, thus $\forall P, \exists S$ such that S is a closed set.

1.3 FORM is the minimum of the set of Closed

First let's prove that FORM is closed. It seems to be trivial that the intersection of closed sets is closed but formal proof is still needed. We denote a set from the intersection construction as T_0.

Lemma 1.4. *T_0 is closed*

We are proving that the set T_0 we get from the intersection construction still satisfy the rules that define what is closed.

Proof. From rule 1,

$$\begin{aligned}
&\forall S \in ACS, P \subseteq S \\
&\quad \to \forall p \in P, \forall S \in ACS, p \in S \\
&\quad \to \forall p \in P, p \in \bigcap ACS \\
&\quad \to P \in T_0
\end{aligned}$$

From rule 2,

$$\begin{aligned}
\forall S \in ACS, T_0 \subseteq S& \\
\text{let } w \in T_0, \forall S \in ACS, w \in S, \neg w \in S& \\
\to \forall w \in T_0, \neg w \in \bigcap ACS& \\
\to \forall w \in T_0, \neg w \in T_0&
\end{aligned}$$

From rule 3,

$$\begin{aligned}
\forall S \in ACS, T_0 \subseteq S& \\
\text{let } w_1, w_2 \in T_0, \forall S \in ACS, w_1 \vee w_2 \in S& \\
\to \forall w_1, w_2 \in T_0, w_1 \vee w_2 \in \bigcap ACS& \\
\to \forall w_1, w_2 \in T_0, w_1 \vee w_2 \in T_0&
\end{aligned}$$

□

Lemma 1.5. *T_0 is **a** smallest Closed set*

Proof. The proof of minimality is trivial. $T_0 = \bigcap ACS \to \forall S \in ACS, T_0 \subseteq S$ □

Lemma 1.6. *T_0 is **the** smallest Closed set (unique). In other word,*

$$\forall T_0' \in \{T \mid \forall S \in ACS, T \subseteq S\}, T_0' = T_0$$

Proof. Suppose otherwise, $\exists T_0' \in ACS \neq T_0$.

$$\begin{aligned}
T_0, T_0' \in ACS &\to \forall S \in ACS, T_0 \subseteq S, T_0' \subseteq S \\
&\to T_0 \subseteq T_0' \wedge T_0' \subseteq T_0 \\
&\to T_0 = T_0'
\end{aligned}$$

A contradiction. □

Now we can use the generative rule to define formula, which is called **Backus-Naur form**. Note that a context-free grammar is a type of formal language. Backus Naur form is a specification language for this type of grammar. It is used to describe language syntax.

1.4 Subformula

Definition 1.7. *ϕ_1 is a subformula of ϕ_2, denoted by*

$$\phi_1 \preccurlyeq \phi_2$$

as a relation is

- *reflexive:* $\forall \phi \in FORM, \phi \preccurlyeq \phi$
- *transitive:* $\forall \phi_1, \phi_2, \phi_3 \in FORM, (\phi_1 \preccurlyeq \phi_2) \wedge (\phi_2 \preccurlyeq \phi_3) \to \phi_1 \preccurlyeq \phi_3$

Inductive definition:

$$\begin{aligned} \forall \phi \in FORM, \phi &\preccurlyeq (\neg\phi) \\ \forall \phi_1, \phi_2 \in FORM, \phi_1 &\preccurlyeq (\phi_1 \vee \phi_2) \\ \phi_2 &\preccurlyeq (\phi_1 \vee \phi_2) \end{aligned}$$

We might intuitively think that subformula can be defined using the notion of substring, since we have been considering formula as concatenation of proposition and connector symbols. However based on out inductive definition, we can come up with the following counter example to a definition using substring: A formula ϕ_1 is a subformula of ϕ_2 if and only if ϕ_1 is a well-formed formula, and is a substring to ϕ_2

$$p \vee q \not\preccurlyeq \neg p \vee q$$

What if we tighten the syntax by strictly enforce the use of parenthese whenver we can

2 Sematics

Definition 2.1. *Sematics describe meanings.*

2.1 Truth Assignment/Proposition evaluation

A truth assignment or a valuation of propositions is a function that maps from the set of propositions to true and false.

$$v : P \to \{T, F\}$$

$$v \in [P \to \{T, F\}]$$

Example 2.2. *Consider $v = \{p_1 \mapsto T, p_2 \mapsto F\}$, we can derive*

$$\begin{aligned} &v(p_1) = T \\ &v(p_2) = F \\ &v(\neg p_2) = T \\ &v(p_1 \vee p_2) = T \text{ etc.} \end{aligned}$$

2.2 Entail/Model

We denote entailment of an evaluation function to formula as

$$v \models \psi$$

We also indeuctively define the following rule

$$\begin{aligned} \forall p \in P, v \models p &\iff v(p) \\ v \models \psi &\iff v \not\models \neg\psi \\ v \models \psi_1 \vee \psi_2 &\iff (v \models \psi_1) \vee (v \models \psi_2) \end{aligned}$$

2.2.1 Valuation and Model

A valuation v is a model of a formula ϕ, if and only if $v(\phi) = T$.

- $\forall \phi$ over $P, \exists 2^{|P|}$ valuations.
- A formula can have many or one or none models

2.2.2 Logical Equivalence

Definition 2.3. *Syntax definition of logical equivalence is as follows:*

$$\phi_1 \equiv \phi_2 \iff \forall v : P \mapsto \{T, F\}(v \models \phi_1), v \models \phi_2$$

Example 2.4.

$$p \vee (\neg p) \equiv q \vee (\neg q)$$

$$\begin{aligned}
&\forall v \models (p \vee (\neg p)),\\
&v \models p \ or \ v \models (\neg p)\\
&v \models p \ or \ v \not\models p\\
&\forall v : P \mapsto \{T, F\},
\end{aligned}$$

TODO

2.3 Satisfiability of Formula

Definition 2.5. *Let $\phi \in FORM, \phi$ is satisfiable*

$$\iff \exists v : P \mapsto \{T, F\}, v \models \phi$$

2.3.1 Valid formula(tautology)

Definition 2.6. *Tautology: ϕ is a tautology $\iff \forall v, v \models \phi$*

Example 2.7.

$$\phi = \neg(p \vee (\neg p))$$

From example 2.4, we have

$$\begin{aligned}
&\forall v, v \models (p \vee (\neg p))\\
&\to \forall v, v \not\models \neg(p \vee (\neg p))
\end{aligned}$$

Thus ϕ is NOT satisfiable.

2.4 Satisfiability of a set of formula

Set Satisfication

Definition 2.8. *Let $\Gamma \subseteq FORM$,*

$$v \models \Gamma \iff \forall \phi \in \Gamma, v \models \phi$$

Set Satisfiability

Definition 2.9. *Let* $\Gamma \subseteq FORM$*,* Γ *is satisfiable*

$$\iff \exists v, \forall \phi \in \Gamma, v \models \phi$$

Example 2.10. $\Gamma = \emptyset$*,* Γ *is satisfiable.*

Proof. Consider $v : P \mapsto \{T, F\}, v(p) = T \forall p \in P$, $\forall \phi \in \Gamma, v \models \phi$ is vacuously true. $\square$

Example 2.11. $\Gamma = FORM$*,* Γ *is NOT satisfiable.*

Proof. Suppose otherwise,

$$\begin{aligned} &\exists v, \forall \phi \in \Gamma, v \models \phi \\ &\forall \phi \in \Gamma, (\neg \phi) \in \Gamma \\ &v \models (\neg \phi) \end{aligned}$$

A contradiction. $\square$

2.4.1 Relevance

Definition 2.12. *Define set* $OCCUR(\phi) \subseteq P$*, which contains propositions appear in* ϕ*. Inductively defined as*

$$\begin{aligned} &OCCUR(\phi) = \{p\} \iff \phi \equiv p, p \in P \\ &OCCUR(\neg \phi) = OCCUR(\phi) \\ &OCCUR(\phi_1 \vee \phi_2) = OCCUR(\phi_1) \cup OCCUR(\phi_2) \end{aligned}$$

Lemma 2.13. *Let* $\phi \in FORM$*, let* v_1, v_2 *be valuations over* P*.*

$$\forall p \in OCCUR(\phi), v_1(p) = v_2(p) \to (v_1 \models \phi \to v_2 \models \phi)$$

v_1, v_2 *can be different valuations over* $p' \notin OCCUR(\phi)$

2.4.2 Logical consequence

Semantic logical consequence:

Definition 2.14.

$$\Gamma \models \phi \iff \forall v \models \Gamma, v \models \phi$$

Syntactic logical consequence: (logical deriability)

Definition 2.15. *A formula* ϕ *is a syntactic consequence of* Γ *if there exists a formal proof (a finite sequence of formulas) ending in* ϕ*, where each step is either a premise from* Γ*, an axiom, or follows from previous steps via an inference rule (like Modus Ponens).*

Material Implication $(\to)$ **vs. Logical Consequence** $(\models)$

- Material implication is a connector within a formula, $p \to q \equiv \neg(p \vee q)$, and can therefore be eval to true or false in a single model.
- Logical consequence is a relation between formulas (meta-logic), it is a statement about all models. $\{p, p \to q\} \models q$

2.5 Modelling Computations as Propositional Logic

Computation problems can be considered as membership check of an encoded problem instance to a language.

$$L = \{p \mid \langle M, p \rangle\}$$

Here instead of arbituary encoding, we consider problem encoding with propositional logics only, and solve with SAT solvers by deciding whether the set of formula we form from any instance of the problem is satisfiable. That is

$$Encode : p \mapsto FORM, \forall p, \exists v \models Encode(p) \iff p \in L$$

The rules of such modelling include:

- Propositional logics only
- The final expression must one well formed fomula
- for problem of size n, the size of the formula ϕ must be in $O(poly(n))$

Note that there can be different definition on the size of the

- problem: in unary, binary etc;
- Formula: Whether by the number of nodes in the syntax tree or length of the string Anything else?

Example 2.16. *Graph coloring problem: given a undirected graph $G = (V, E)$, whether there exists a k coloring of all vertices such that no neighbouring vertices have the same color.*
Consider the following construction of a proposition set P:

$$P = \{p_{v,i} \mid v \in V, i \in [1, k]\}$$

The valuation of proposition $p_{v,i}$ conceptually represents vertex v is assigned color i.
We can formulate constriants as well formed formula, and use conjunction to connect the formula to model satisfaction of multiple constraints.

Corollary 2.17. $\forall \phi_1, \phi_2 \in FORM, (\phi_1 \land \pi_2) \in FORM$

First let us list down the constraints of this problems, including implicit ones in plain words

1. *Neighbouring vertices have different colors.*
2. *All vertices as assigned with exactly one color.*

1. ϕ_{nbr}

First lets formula the constraint between any 2 vertices,

$$\forall (\alpha, \beta) \in E, c(v_\alpha) \neq c(v_\beta)$$

And from our proposition set, $c(v_\gamma) = i \iff \exists v, v \models p_{\gamma,i}$. As such we have

$$\phi_{\alpha,\beta,i} = \neg(p_{\alpha,i} \land p_{\beta,i})$$

This models vertex α and vertex β are not colored with color i at the same time. For every color this needs to hold, thus we have

$$\phi_{nbr(\alpha,\beta)} = \bigwedge_{i \in [1,k]} \phi_{\alpha,\beta,i}$$

For every pair of neighbouring vertices this also needs to hold. Therefore we have

$$\phi_{nbr} = \bigwedge_{(\alpha,\beta) \in E} \phi_{nbr(\alpha,\beta)}$$

2. $\phi_{exactly-1}$

To model exactly 1 coloring on each vertex, we can formulate $\phi_{\geq 1}$ and $\phi_{\leq 1}$ first and use conjunction to connect them. Given our propositions, to make sure vertex v is at least assign a color ($\phi_{v,\geq 1}$),

$$\phi_{v,\geq 1} = \bigvee_{i \in [1,k]} p_{v,i}$$

This needs to hold for every vertex, thus we have

$$\phi_{\geq 1} = \bigwedge_{v \in V} \phi_{v,\geq 1}$$

Then we need to model at most 1, (or less than 2) coloring on each vertex. Vertex v is not assignment with both color i and j can be formulated as

$$\phi_{v,i,j,\leq 1} = \neg(p_{v,i} \wedge p_{v,j})$$

This needs to hold for every distinct pair of colors, thus

$$\phi_{v,\leq 1} = \bigwedge_{i \neq j \in [1,k]} \phi_{v,i,j,\leq 1}$$

For all vertices

$$\phi_{\leq 1} = \bigwedge_{v \in V} \phi_{v,\leq 1}$$

Finally we have

$$\phi_{exactly-1} = \phi_{\leq 1} \wedge \phi_{\geq 1}$$

And the overall formula for an instance of $G = (V, E)$ becomes

$$\phi = \phi_{nbr} \wedge \phi_{exactly-1}$$

Similar to reduction, we need also now to show that

$$\exists v : P \mapsto \{T, F\}, v \models \phi \iff \langle G = (V, E), k \rangle \in L_{color}$$

TODO

3 Compactness Theorem

Definition 3.1. *Axiom of Choice: formally*

$$\forall X, \emptyset \in X \vee \exists f, dom(f) = X \wedge \forall A \in X, f(A) \in A$$

A choice function (or selection) f defined on a collection of nonempty sets X, such that for every set A in the collection X, $f(A)$ is an element in A. That is, Axiom—For any set X of nonempty sets, there exists a choice function f that is defined on X and maps each set of X to an element of that set.

Theorem 3.2. *If P is finite, $FORM$ over P is finite.*

Theorem 3.3. *If P is countable, $FORM$ over P is countable.*

Proof. Given $\forall \phi \in FORM, \phi$ is finitely long, this theorem is another way of stating: The language formed from countable alphabet $P \cup C$ with the inductive construction rules, is countable. We can prove a stronger theorem first. ☐

Theorem 3.4. *All languages formed from a countable alphabet is countable.*

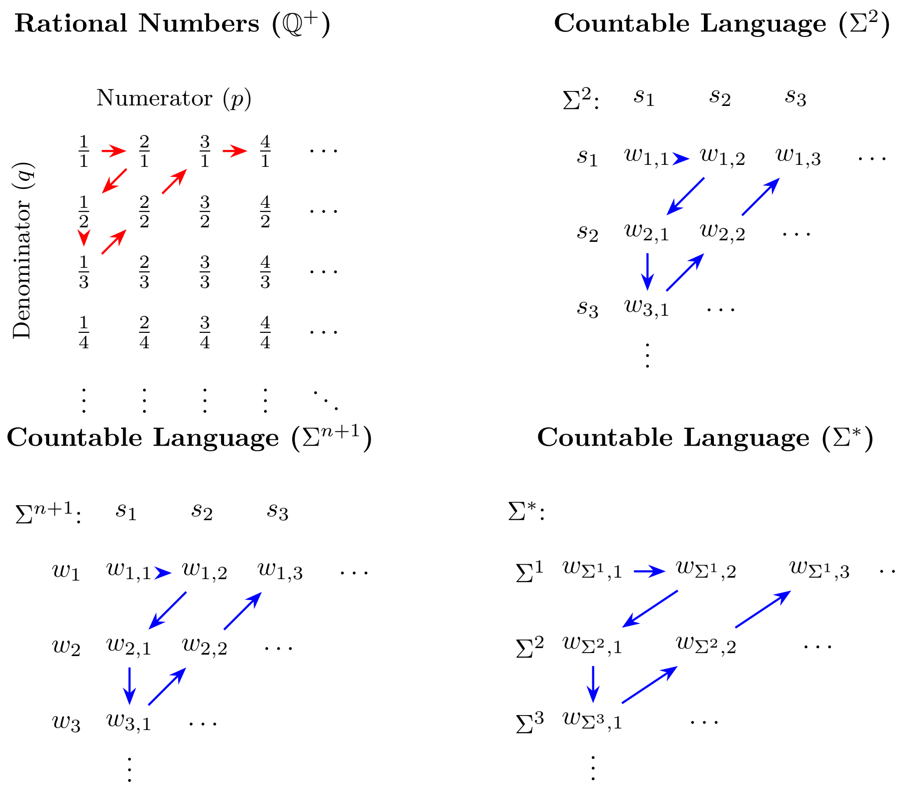

Figure 1: Comparison of diagonal enumeration for numbers vs. strings.

Proof. Consider $L = \Sigma^*$, we will prove that Σ is countable $\to L$ is countable. We use the same diagnol proof for the countability of $\mathbb{Q}$ to show an enumaration of all strings formable from a countable alphabet. To formalize, we use induction to prove that $\forall i \geq 0, \Sigma^i$ is countable
Base case: $\Sigma^1 = \Sigma$, countable as defined.
Inductive hypothesis: Σ^n is countable: we enumerate all strings in Σ^n, and use the same diagonal proof to show the concatenation of one symbol from the countable alphabet to every string Σ^{n+1} is also countable. Thus we can give an enumeration of all Σ^i and use the diaganol argument again to show the existence of an enumeration of Σ^*.
Alternatively, Given

$$\Sigma^* = \bigcup_{i \geq 0} \Sigma^i$$

From the Axiom of Choice, which implies the union of countably many countable set is countable, Σ^* is countable. Since $\forall L$ over $\Sigma, L \subseteq \Sigma^*, \Sigma^*$ is countable, all languages over countable Σ is countable. $\square$

Definition 3.5 (Finitely Satisfiable)**.** *A set of formulas Γ is said to be **finitely satisfiable** if and only if for all finite subsets $\Gamma_0 \subseteq \Gamma$, Γ_0 is satisfiable.*

Theorem 3.6 (Compactness Theorem)**.**

$$\Gamma \textit{ is satisfiable} \iff \Gamma \textit{ is finitely satisfiable}$$

Proof. Consider a special case where Γ is "T-good", defined as:

$$\forall p \in P, (p \in \Gamma \vee \neg p \in \Gamma)$$

Claim: If Γ is T-good, then Γ is finitely satisfiable $\implies \Gamma$ is satisfiable.
Suppose Γ is T-good and finitely satisfiable. We define a candidate valuation v_L such that for all $p \in P$:

$$v_L(p) = \begin{cases} T & \text{if } p \in \Gamma \\ F & \text{if } \neg p \in \Gamma \end{cases}$$

First, we prove that for every $p \in P$, **exactly one** of $\{p, \neg p\}$ is in Γ. By the T-good property, at least one is in Γ. Suppose both $\{p, \neg p\} \subseteq \Gamma$. Consider the finite subset:

$$\Gamma' = \{p, \neg p\}$$

Since Γ' is a finite subset of a finitely satisfiable set, it must be satisfiable. However, $\neg \exists v$ such that $v \models \{p, \neg p\}$, which is a contradiction. Thus, v_L is well-defined.

Now we show $v_L \models \Gamma$. Suppose $\exists \phi \in \Gamma$ such that $v_L \not\models \phi$. Define the following finite sets based on the atoms occurring in ϕ:

$$\begin{aligned} \Gamma_\phi^+ &= \{p \mid p \in \text{OCCUR}(\phi) \wedge v_L(p) = T\} \\ \Gamma_\phi^- &= \{\neg p \mid p \in \text{OCCUR}(\phi) \wedge v_L(p) = F\} \\ \Gamma_\phi &= \{\phi\} \cup \Gamma_\phi^+ \cup \Gamma_\phi^- \end{aligned}$$

Since Γ_ϕ is a finite subset of Γ, it must be satisfiable by some valuation v. For this v:

- $\forall p$ such that $p \in \Gamma_\phi^+$, $v(p) = T$.
- $\forall \neg p$ such that $\neg p \in \Gamma_\phi^-$, $v(\neg p) = T \implies v(p) = F$.

By construction, v and v_L agree on all atoms in $\text{OCCUR}(\phi)$. By the **Relevance Lemma**, $v(\phi) = v_L(\phi)$. Since $v \models \Gamma_\phi$, we have $v \models \phi$. This implies $v_L \models \phi$, contradicting our assumption $v_L \not\models \phi$. Therefore, $\forall \phi \in \Gamma, v_L \models \phi$, which implies $v_L \models \Gamma$, Γ is satisfiable.

Now we will generalize this to all Γ which may not be 'T-good', by showing that we can construct a 'T-good' finitely satisfiable set Δ, from any finitely satisfiable Γ, and therefore show that $\Gamma SAT \iff \Delta SAT$.
We will only prove the case for countable P now. Let an enumeration of $p \in P$ be $\{p_1, p_2, \dots\}$
Define finite subset Γ_i of Γ as such:

$$\begin{aligned} \Gamma_0 &= \Gamma \\ \Gamma_{i+1} &= \begin{cases} \Gamma_i \cup \{p_i\} \text{ if } \Gamma_i \cup \{p_i\} \text{ is finitely satisfiable} \\ \Gamma_i \cup \{\neg(p_i)\} \text{ Otherwise} \end{cases} \end{aligned}$$

Here we need to prove that $\forall i$, between $\Gamma_i \cup \{p_i\}$ and $\Gamma_i \cup \{(\neg p_i)\}$ at least one case is also finitely satisfiable.

Theorem 3.7. *Let Γ be finitely satisfiable set, and ϕ be a formula, then $\Gamma \cup \{\phi\}$ is finitely satisfiable or $\Gamma \cup \{(\neg\phi)\}$ is finitely satisfiable.*

Proof. We prove the contrapositive of this statement: if $\Gamma \cup \{\phi\}$ and $\Gamma \cup \{(\neg\phi)\}$ are both not finitely satisfiable, Γ is not fintely satisfiable. Given both are not finitely satisfiable, there exist finite subsets $\Gamma_1, \Gamma_2 \subseteq \Gamma$ such that $\Gamma_1 \cup \{\phi\}$ and $\Gamma_2 \cup \{(\neg\phi)\}$ are not satisfiable. Therefore for all valuations, it must not model either Γ_1 or $\{\phi\}$, and not model either Γ_2 or $\{(\neg\phi)\}$. Denote the set of valuations that models formula ψ as $models(\psi)$, the set of valuations that models set of formula Σ as $models(\Sigma)$

$$\begin{aligned}
models(\Gamma_1) \cap models(\phi) &= \emptyset \\
&\rightarrow \forall v \in models(\Gamma_1), v \not\models \phi \rightarrow v \models (\neg\phi) \\
&\rightarrow models(\Gamma_1) \subseteq models((\neg\phi)) \\
models(\Gamma_2) \cap models((\neg\phi)) &= \emptyset \\
&\rightarrow \forall v \in models(\Gamma_2), v \not\models (\neg\phi) \rightarrow v \models (\phi) \\
&\rightarrow models(\Gamma_2) \subseteq models(\phi)
\end{aligned}$$

Now consider the set of valuations that models $\Gamma_1 \cup \Gamma_2$

$$\begin{aligned}
&models(\Gamma_1 \cup \Gamma_2) = models(\Gamma_1) \cap models(\Gamma_2) \\
&(models(\Gamma_1) \cap models(\Gamma_2)) \subseteq (models(\phi) \cap models((\neg\phi))) = \emptyset
\end{aligned}$$

Since $\Gamma_1 \cup \Gamma_2$ is a finite subset of Γ that is not satisfiable, Γ is not finitely satisfiable. □

Now prove that

Lemma 3.8. *$\forall i, \Gamma_i$ is finitely satisfiable.*

Proof. Base case:

$$\Gamma_0 = \Gamma \text{ is finitely satisfiable by definition}$$

Inducetive hypothesis: Suppose

$$\Gamma_n \text{ is finitely satisfiable}$$

By Lemma 3.7, $\Gamma_n \cup \{p_n\}$ and $\Gamma_n \cup \{(\neg p_n)\}$ there must be at least one that is finitely satisfiable. $\rightarrow \Gamma_{n+1}$ is therefore finitely satisfiable. □

Next we will show that

Lemma 3.9. *$\Gamma_U = \bigcup_{i\geq 0} \Gamma_i$ is finitely satisfiable.*

Proof. Consider finite set $\Gamma' \subseteq \Gamma_U$, since Γ' is finite, and $\forall k \geq 0, \Gamma_k \subseteq \Gamma_{k+1}$ by definition, $\exists j \geq 0, \Gamma' \subseteq \Gamma_j = \bigcup_{i\geq 0}^{j} \Gamma_i$. By Lemma 3.8, Γ_j is satisfiable $\rightarrow \Gamma'$ is finitely satisfiable. □

Lemma 3.10. *Γ_U is satisfiable.*

Proof. From the construction, $\forall i \geq 0$ at least one of p_i and $(\neg p_i)$ is in $\bigcup_{i\geq 0} \Gamma_i$. Since $\bigcup_{i\geq 0} \Gamma_i$ is finitely satisfiable, $\forall i \geq 0, p_i$ and $(\neg p_i)$ cannot both be in $\bigcup_{i\geq 0} \Gamma_i$. Otherwise $\{p_i\} \cup \{(\neg p_1)\} \subseteq \bigcup_{i\geq 0} \Gamma_i$ is not satisfiable, contradiction. □

Now we have proven that the construction proposed from a finitely satisfiable formula set Γ to $\bigcup_{i\geq 0} \Gamma_i$ is also finitely satisfiable. Now to complete the proof of the compactness theorem, we need to prove that $\exists v, v \models \Gamma$.

Proof. From Lemma 3.10, $\forall i \geq 0, (p_i \in \Gamma_U) \oplus ((\neg p_i) \in \Gamma_U)$, $\exists v, v \models \Gamma_U$

$$v(p) : P \mapsto \{T, F\} = \begin{cases} T, \text{ if } p \in \Gamma_U \\ F, \text{ if } (\neg p) \in \Gamma_U \end{cases}$$

Define the literal set $Literals = \{p_i, (\neg p_i) \mid i \geq 0\}$, and function

$$\tau(p) : P \mapsto Literals = \begin{cases} p, \text{ if } p \in \Gamma_U \\ (\neg p), \text{ if } (\neg p) \in \Gamma_U \end{cases}$$

Lemma 3.11.

$$v \models \Gamma_U \to v \models \tau(\Gamma)$$

Let $\phi \in \Gamma$, Define literal set $X = \{\tau(p) \mid p \in OCCUR(\phi)\}$ where X is a set containing single proposition literals. By definition we have

$$X \subseteq \Gamma_U \to v \models X \to \forall \psi \in X, v(\psi) = T$$

and $OCCUR(X) = OCCUR(\phi)$ Since both X is finite, $X \cup \{\phi\}$ is a finite subset to Γ_U, thus satisfiable. $\to \exists v', v' \models X \cup \{\phi\} \to v' \models X, v' \models \phi \to \forall \psi \in X, v'(\psi) = T$.

$$\forall p \in OCCUR(\phi), v(p) = v'(p), v' \models \phi \to v \models \phi$$

Therefore, the valuation that models enlarged construction Γ_U also models Γ, thus Γ is satisfiable. ☐

☐

3.1 Application of the Compactness theorem for proposition logics

We have already shown that we can use propositional logic to model computational problems, the compactness theorem allows us to extend problem modelling to instances of infinite size. Recall the k-colorable problem, where we used propositional logic to model any finite size instance.

Theorem 3.12. *A graph is k-colorable iff every finite subgraph is k-colorable.*

Definition 3.13 (Infinite graph)**.** *$G = (V, E)$ is an infinite graph if V is infinite.*

Definition 3.14 (Subgraph)**.** *$G' = (V', E')$ is a subgraph to G if $V' \subseteq V, E' \subseteq E$*

Definition 3.15 (Induced subgraph)**.** *$G' = (V', E')$ is an induced subgraph to G if G' is a subgraph to G, and $\{u, v\} \in E' \iff \{u, v\} \in E \wedge u, v \in V'$*

Proof. Since we have formulated any finite instance of k-colorable graph as propositional logic formula, we can define infinite set

$$\Gamma_G = \{\phi_{G'}^{k-colorable} \mid G' \text{ is finite subgraph of } G\}$$

We will first prove that

Lemma 3.16. *G is finitely k-colorable $\to \Gamma_G$ is finitely satisfiable.*

Proof. We will reuse the proposition definition from last chapter, where $|V|$ and thus P is countable

$$P = \{p_{v,i} \mid v \in V, i \in [1, k]\}$$

For each instance on $G = (V, E)$ and k colors, we formulate as such to enforce constraints on edges (No same color on adjacent vertices), and vertices (exactly 1 coloring on each vertex).

$$\phi = \phi^{nbr} \wedge \phi^{exactly-1-color}$$

Consider the infinite graph $G = (V, E)$

$$\begin{aligned}
\Gamma_G &= \Gamma_G^{nbr} \cup \Gamma_G^{exactly-1-color} \\
\Gamma_G^{exactly-1-color} &= \{\phi_v^{exactly-1-color} \mid v \in V\} \\
\phi_v^{exactly-1-color} &= (\bigvee_{i \in [1,k]} p_{v,i}) \wedge (\bigwedge_{i \neq j \in [1,k]} (p_{v,i} \to p_{v,j})) \\
\Gamma_G^{nbr} &= \{\phi_{\{u,v\}}^{nbr} \mid \{u, v\} \in E\} \\
\phi_{\{u,v\}}^{nbr} &= \bigwedge_{i \in [1,k]} (p_{u,i} \to p_{v,i})
\end{aligned}$$

Suppose the contrary, where $G = (V, E)$ is finitel k-colorable, Γ_G is not finitely satisfiable, $\exists$ finite subset $\Gamma_0 \subseteq \Gamma_G, \Gamma_0$ is not satisfiable. Let $G_0 = (V_0, E_0)$ be a finite subgraph of G, such that

$$V_0 = \{v \mid v \in V, \phi_v^{exactly-1-color} \in \Gamma_0\} \cup \{u \mid u \in V, \exists u', \phi_{\{u,u'\}}^{nbr} \in \Gamma_0\}$$

$$E_0 = \{\{u, v\} \mid \{u, v\} \in E, u, v \in V_0\}$$

Here we are formulating an instance of finite k-colorable induced subgraph of G from finite subset Γ_0. Now we need to show that G_0 is not k-colorable to reach the contradiction, with another contradiction.
Suppose G_0 is k-colorable, let C_0 be a k-coloring of G_0, define valuation

$$v_0(p_{u,i}) = \begin{cases} T, & \text{if } C_0(u) = i \\ F, & \text{if } C_0(u) \neq i \end{cases}$$

Let

$$\Gamma_0' = \{\phi_v^{exactly-1-color} \mid v \in V_0\} \cup \{\phi_{\{u,v\}}^{nbr} \mid \{u, v\} \in E_0\}$$

Consider

$$\Gamma_0 \subseteq \Gamma_0'$$

Since Γ_0' is the same construction from k-colorability to formuala, G_0 is k-colorable, we have

$$v_0 \models \Gamma_0'$$

By relevance lemma

$$v_0 \models \Gamma_0$$

Contradicting to our initial assumption that Γ_0 is NOT satisfiable. □

By Compactness theorem, Γ_G is finitely satisfiable $\rightarrow \Gamma_G$ is satisfiable.

Lemma 3.17. *Γ_G is satisfiable $\rightarrow G$ is k-colorable*

Proof. Γ_G is satisfiable $\rightarrow \exists v, v \models \Gamma_G$. We can construct a coloring C from v similarly to how we constructed v_0 from C_0 from previous proof:

$$C(v) = i, \text{ if } v(p_{v,i}) = T$$

By construction the coloring is valid, detailed technicality is ommitted. □

□

4 Proof System

4.1 Hilbert Style Proof System

In HSPS we deal with formulas formed over

$$\phi + =$$

Four axiom schema of the hilberts style proof system. Axiom schema means that the ϕ, ψ, ρ in these schema are placehoders that can be replaced by any valid formula

1. Axiom 1: Simplification

$$\frac{}{p \rightarrow (q \rightarrow p)}$$

2. Axiom 2: Distribution

$$\frac{}{(p \to (q \to r)) \to ((p \to q) \to (p \to r))}$$

3. Axiom 3: Double Negation Elimination (DNE)

$$\frac{}{((p \to \bot) \to \bot) \to p}$$

4. Rule of Inference: Modus Ponens

$$\frac{p \to q \qquad p}{q}$$

The horizontal lines are judgement lines, which means that whats below can be inferred from whats above. The first 3 rules has no assumptions or premises, meaning they are ways we can introduce arbituary claims or assumption into our proofs.

Definition 4.1 (Proof)**.** *A proof of the consequence of ϕ from Γ is a finite sequence of formula $c_1, c_2, \ldots, c_k$ where $\forall i \leq k, c_i \in \Gamma$ or, c_i is obtained using one of the 4 axioms:*

- $\exists j, l < i, c_j \equiv c_l \to c_i$
- $\exists \phi, \psi, c_i \equiv \phi \to (\psi \to \phi)$
- $\exists \phi, \psi, \rho, c_i \equiv [\phi \to (\psi \to \rho)] \to [(\phi \to \psi) \to (\phi \to \rho)]$
- $\exists \phi, c_i \equiv ((\phi \to \bot) \to \bot) \to \phi$

Definition 4.2 (Derivable)**.** *ϕ is derivable from Γ if there is a proof in HSPS of "ϕ is a consequence of Γ"*

$$\Gamma \vdash \phi$$

Example 4.3.

$$\Gamma = \{p, p \to q\}, \quad \phi = q$$

Proof. We can construct proof sequence:

$$\begin{aligned} &c_1 = p && (p \in \Gamma) \\ &c_2 = p \to q && (p \to q \in \Gamma) \\ &c_3 = q && (\text{Modus Ponens on } c_1, c_2) \end{aligned}$$

□

Sometimes HSPS requires us to creatively introduce new hypothesis to assist in the proof.

Example 4.4.

$$\Gamma = \{\bot\}, \phi = p$$

Intuitively this does not sound provable, just from $\bot$, but axioms that require no premises allows us to introduce helper hypothesis. In this case $(p \to \bot)$

Proof.

$$\begin{aligned} &c_1 = \bot && (\bot \in \Gamma) \\ &c_2 = \bot \to [(p \to \bot) \to \bot] && (\text{Axiom 1: Simplification}) \\ &c_3 = (p \to \bot) \to \bot && (\text{Modus Ponens on } \bot, (p \to \bot)) \\ &c_4 = [(p \to \bot) \to \bot] \to p && (\text{Axiom 3: Double negation}) \\ &c_5 = p && (\text{Modus Ponens on } c_3 \equiv [(p \to \bot) \to \bot], c_4 \equiv ([(p \to \bot) \to \bot] \to p)) \end{aligned}$$

□

As seen from this example, HSPS can get convoluted and unintuitive for relatively straightforward consequences.

Theorem 4.5 (Soundness of HSPS)**.** *HSPS is sound.*

$$\Gamma \vdash \phi \implies \Gamma \models \phi$$

Proof. Define property $H(i) : \forall j \leq i, \Gamma \models c_j$.
Base case: $i = 1$. Suppose $\Gamma \vdash \phi, \exists \pi = c_1, c_2, \ldots, c_k, c_k = \phi$. By definition, $c_1 \in \Gamma$ or

- $\exists \phi, \psi, c_1 \equiv \phi \to (\psi \to \phi)$
- $\exists \phi, \psi, \rho, c_1 \equiv [\phi \to (\psi \to \rho)] \to [(\phi \to \psi) \to (\phi \to \rho)]$
- $\exists \phi, c_1 \equiv ((\phi \to \bot) \to \bot) \to \phi$

Note that Modus Ponens is not possible to yield c_1 as we need at least 1 previous formula. If $c_1 \in \Gamma, \Gamma \models c_1$; Else, if c_1 comes from any other axiom, c_1 is valid, i.e. $\forall v, v \models c_1$, therefore $\Gamma \models c_1$

□

Theorem 4.6 (Completeness of HSPS)**.** *HSPS is complete.*

$$\Gamma \models \phi \implies \Gamma \vdash \phi$$

Proof. □

4.2 Resolution Proof System

Definition 4.7 (Clause)**.** *For each proposition p define literals p and $\neg p$. A clause C is finite set of literals over the proposition set P.*

$$v \models C = \{\ell_1, \ell_2, \ldots, \ell_k\} \iff \exists j, v \models \ell_j$$

As such, a cluase C is satisfiable if $\exists v, v \models C$

Note that from this defintion, an empty clause (an empty set of literals) is not satisfiable, as intuitively, a valuation models a clause if there exists at least one literal in the clause such that $v \models \ell$.

Definition 4.8 (Conjunctive normal form (CNF))**.** *A formula ϕ is said to be in Conjunctive Normal Form is*

$$\phi \equiv \bigwedge_{i=1}^{k} C_i$$

Definition 4.9 (Satisfiability of a set of clauses)**.**

$$\Gamma = \{C_1, C_2, \ldots\}$$

- Γ *is satisfiable if* $\exists v, \forall C \in \Gamma, v \models C$
- Γ *is unsatisifiable if* $\forall v, \exists C \in \Gamma, v \not\models C$

By this definition,

- $\Gamma = \emptyset$ is vacuously satisfiable
- $\Gamma = \{\emptyset\}$ is no satisifiale.
- Any set of clauses that contains the empty set is unsatisfiable

Tutorial

Definition 4.10 (Generic Closure)**.** *Let U be a universe of elements and let O be a set of functions whose domain is U and codomain is $\mathscr{P}(U)$. The arity of a function $f \in O$ is denoted by arity(f). A function f of arity r has type*

$$f : U^r \to \mathscr{P}(U).$$

*Let $I \subseteq U$ be an initial set. A set $S \subseteq U$ is said to be (I, O)-*closed *if:*

1. *$I \subseteq S$.*
2. *For every $f \in O$ with arity$(f) = r$ and for all $e_1, \dots, e_r \in U$, if $e_1, \dots, e_r \in S$, then $f(e_1, \dots, e_r) \subseteq S$.*

Lemma 4.11. *Let U be a universe, O a set of operators, and $I \subseteq U$ an initial set. There exists a smallest (I, O)-closed set.*

Define $\mathcal{P}(U)$ as a collection of all ordering of all elemenst in the power set $\mathscr{P}(U)$

Q1

Claim: The smallest (I, O)-closed set can be constructed by taking the intersection of all (I,O)-closed set, ie.

$$AIOCS = \{S \subseteq U \mid S \text{ S is (I,O)-closed}\}$$
$$T_0 = \bigcap AIOCS$$

Existence

First we prove the existence of such a set.

Proof. Suppose otherwise,

$$T_0 = \bigcap AIOCS = \emptyset$$
$$T_0 = \{x \in \mathcal{P}(U) \mid \forall S \in AIOCS, x \in S\}$$
$$T_0 = \emptyset \to x \in S \text{ is vacuously true } \forall x \in \mathcal{P}(U) \to T_0 = \mathcal{P}(U)$$

contradiction. □

Closed

Now we prove that T_0 is (I,O)-closed.

Proof. Property 1:

$$\begin{aligned} &\forall S \in AIOCS, I \subseteq S \\ &\quad \to \forall i \in I, \forall S \in AIOCS, i \in S \\ &\quad \to \forall i \in I, i \in \bigcap AIOCS \\ &\quad \to \forall i \in I, i \in T_0 \end{aligned}$$

Property 2:

$$\begin{aligned} &\forall S \in AIOCS, T_0 \subseteq S \\ &\quad \text{Let } e_1, e_2, \dots, e_r \in T_0, \forall S \in AIOCS, e_1, e_2, \dots, e_r \in S \\ &\quad \to \exists f \in \mathcal{O}, f(e_1, e_2, \dots, e_r) \in S \\ &\quad \to \forall e_1, e_2, \dots, e_r \in T_0, \exists f \in \mathcal{O}, f(e_1, e_2, \dots, e_r) \in \bigcap AIOCS \\ &\quad \to f(e_1, e_2, \dots, e_r) \in T_0 \end{aligned}$$

□

Uniqueness

Claim:

$$\forall T_0' \in \{T \in \mathcal{P}(U) \mid \forall S \in AIOCS, T \subseteq S\}, T_0' = T_0$$

Proof. Suppose otherwise, $\exists T_0' \in AIOCS, T_0' \neq T_0$.

$$\begin{aligned} T_0, T_0' \in AIOCS &\rightarrow \forall S \in AIOCS, T_0 \subseteq S, T_0' \subseteq S \\ &\rightarrow T_0 \subseteq T_0', T_0' \subseteq T_0 \\ &\rightarrow T_0 = T_0' \end{aligned}$$

□

Q2

For each of the following strings, prove or disprove that it is a well-formed formula:

1. $w_1 = \epsilon$ (the empty string).
2. $w_2 = (\neg p)$ where $p \in P$.
3. $w_3 = (p\neg)$ where $p \in P$.

2.1

$$w_1 = \epsilon \notin FORM$$

Claim: $\forall w \in FORM, |w| > 0$ I'd like to use definition 2.19 and lemma 2.21 which I proved later without using the conslusion from this question.

$$FORM = INDFORM = \bigcup_{i \geq 0} FORM_i$$

Proof. Base case:

$$\begin{aligned} &FORM_0 = P \\ &\forall \phi \in FORM_0, \phi = p \in P \\ &\rightarrow \forall \phi \in FORM_i, |\phi| > 0 \end{aligned}$$

Inductive Hypothesis: Suppose

$$\forall i \geq 0, \forall \phi \in FORM_i, |\phi| > 0$$

$$\begin{aligned} &FORM_{i+1} = FORM_i \cup \{(\neg \phi) \mid \phi \in FORM_i\} \cup \{(\phi_1 \vee \phi_2) \mid \phi_1, \phi_2 \in FORM_i\} \\ &\forall \psi \in FORM_{i+1}, \exists \phi \in FORM_i, |\psi| = |\phi| \text{ or } |\psi| = |\phi| + 3 \text{ or} \\ &\exists \phi_1, \phi_2 \in FORM_i, |\psi| = |\phi_1| + |\phi_2| + 3 \end{aligned}$$

□

2.2

$$p \in P \rightarrow p \in FORM \rightarrow (\neg p) \in FORM$$

2.3

$(p\neg) \notin FORM$

Proof. Suppose otherwise, $(p\neg\epsilon) \in FORM$

$$\begin{aligned} \forall \phi \in FORM, (\neg\phi) \in FORM \\ \to \epsilon \in FORM \end{aligned}$$

Contradiction from 2.1 □

Q3

Fix $U = \mathbb{R}$. For each of the following sets S, define a finite initial set $I \subseteq \mathbb{R}$ and a set of operators O with $|O| < 10$ such that the smallest (I, O)-closed set is S.

1. $S = \{0, 1, 2, 3, \dots\}$.
2. $S = \{1, 3, 5, 7, \dots\}$.
3. $S = \mathbb{Q}$.
4. $S = \left\{\frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \frac{5}{3}, \frac{7}{3}, \dots\right\}$.

Answer:

1. $\mathbb{N} : I = \{0\}, O = \{f(x) = x + 1\}$
2. $Odd : I = \{1\}, O = \{f(x) = x + 2\}$
3. $\mathbb{Q} : I = \{0\}, O = \{f(x) = -x, f(x) = x + 1, f(x, y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ \emptyset & \text{if } y = 0 \end{cases}\}$
4. $\{\frac{x}{3} \mid x \in \mathbb{N}\} \setminus \mathbb{N} : I = \{\frac{1}{3}, \frac{2}{3}\}, O = \{f(x) = x + 1\}$

Finite Directed Graphs

Definition 4.12 (Graph Reachability)**.** *Let $G = (V, E)$ be a directed graph. A path from s to t is a sequence of vertices $\pi = u_1, \dots, u_k$ such that:*

1. *$u_1 = s$,*
2. *$(u_i, u_{i+1}) \in E$ for all $1 \leq i < k$,*
3. *$u_k = t$.*

The length of π is $k - 1$. We say that t is reachable from s if $s = t$ or such a path exists. The set of ancestors of t is defined as

$$Reach_t = \{v \in V \mid t \text{ is reachable from } v\}.$$

Q4

Let $G = (V, E)$ be a directed graph and let $t \in V$. Fix the universe $U = V$.

1. Define an initial set I (with $|I| \leq 2$) and a unary operator f such that the smallest $(I, \{f\})$-closed set is the set of vertices reachable from t.
2. Let $d \in \mathbb{N} \setminus \{0\}$ and $r < d$. Define I and a unary operator f such that the smallest $(I, \{f\})$-closed set is the set of vertices reachable from s via paths of length ℓ with $\ell \equiv r \pmod d$.

4.1

$I = \{t\}, O = \{f(v) = succ(v)\}$

4.2

$I = \{s\} \cup \{v \in V \mid v \in succ^r(s)\}, O = \{f(v) = succ^d(v)\}$

Q5

Let $G = (V, E)$ be a directed graph and fix $U = V \times V$. Define an initial set I with $|I| \leq |V|$ and a binary operator f such that the smallest $(I, \{f\})$-closed set is

$$AllReachPairs = \{(u, v) \mid v \text{ is reachable from } u\}.$$

5.1

$I = \{(v, v) \mid v \in V\}, O = \{f((v_1, v_2), (v_3, v_4)) \mapsto \begin{cases} \{(v_1, v_4)\} & \text{if } v_2 = v_3 \\ \emptyset & \text{if } v_2 \neq v_3 \end{cases}\}$

Inductive Definitions

Definition 4.13 (Inductive Well-Formed Formulae)**.** *Let P be a set of propositions and $C = \{\neg, \vee\}$. Define sets $FORM_i$ as follows:*

$$FORM_0 = P,$$
$$FORM_{i+1} = FORM_i \cup \{(\neg w) \mid w \in FORM_i\} \cup \{(w_1 \vee w_2) \mid w_1, w_2 \in FORM_i\}.$$

Define

$$INDFORM = \bigcup_{i \geq 0} FORM_i.$$

Q6

1. Show that $FORM_i \subseteq FORM_{i+1}$ for all i.
2. Prove or disprove that $FORM_i = FORM_{i+1}$ for some i.
3. Prove or disprove that $FORM_i$ is (P, C)-closed for all i.
4. Prove or disprove that $FORM_i$ is (P, C)-closed for some i.

6.1

Base case: $FORM_0 \subseteq FORM_1$ as

$$FORM_1 = FORM_0 \cup \{(\neg w) \mid w \in FORM_0\} \cup \{(w_1 \vee w_2) \mid w_1, w_2 \in FORM_0\}$$

Wait this is just trivial

6.2

Claim:

$$\exists i \geq 0, FORM_i = FORM_{i+1}$$

It is obvious that $\forall i \geq 0, FORM_i$ is finite. Suppose

6.3

Claim:

$$\forall i \geq 0, FORM_i \text{ is (P,C)-closed}$$

Proof. Suppose otherwise, consider $FORM_0$,

$$FORM_0 = P$$
$$\forall p \in P, (\neg p) \in FORM_0$$

Contradiction □

6.4

Claim:

$$\exists i \geq 0, FORM_i \text{ is (P,C)-closed}$$

Proof.

Lemma 4.14.

$$\forall i \geq 0, \forall \phi \in FORM_i, |\phi| \leq 2^{i+3} - 3$$

The longest formula in $FORM_i$ has size $2^{i+3} - 3$. Parentheses are counted in string length

Proof. Base case:

$$FORM_0 = P$$
$$\forall \phi \in FORM_0, |\phi| = 1$$

Inductive hypothesis:

$$\forall i \geq 0, \forall \phi \in FORM_i, |\phi| \leq 2^{i+3} - 3$$

Consider all new strings formed from the longest string $\phi \in FORM_i$,

$$|\phi| = 2^{i+3} - 3$$
$$|(\neg\phi)| = 2^{i+3}$$
$$|(\phi \vee \phi)| = 2 \times (2^{i+3} - 3) + 3 = 2^{(i+1)+3} - 3$$

□

Suppose $\exists i \geq 0, FORM_i$ is (P,C)-closed, and from lemma 2.20 the longest formula in $FORM_i$ has size $|\phi_{max}| = 2^{i+3} - 3$. $\rightarrow \forall \phi \in FORM_i, (\neg\phi) \in FORM_i \rightarrow \phi'_{max} = (\neg\phi_{max}) \in FORM_i, |\phi'_{max}| > 2^{i+3} - 3$, a contradiction.

□

Lemma 4.15. *$FORM = INDFORM$.*

Q7

Prove the lemma by showing:

1. $INDFORM$ is (P, C)-closed.
2. For every (P, C)-closed set S, $INDFORM \subseteq S$.

7.1

Claim: $INDFORM$ is (P,C)-closed

Proof. **Claim:** $P \subseteq INDFORM$

First condition of (P,C)-closed set:

$$FORM_0 \subseteq INDFORM \to P \subseteq INDFORM$$

Claim: $\forall w \in INDFORM, (\neg w) \in INDFORM$

$$\begin{aligned}
INDFORM = \bigcup_{i \geq 0} FORM_i & \\
&\to \forall i \geq 0, FORM_i, FORM_{i+1} \in INDFORM \\
&\quad \forall w \in INDFORM, \exists j \geq 0, w \in FORM_j \\
&\to (\neg w) \in FORM_{j+1} \to (\neg w) \in INDFORM
\end{aligned}$$

Claim: $\forall w_1, w_2 \in INDFORM, (w_1 \vee w_2) \in INDFORM$

Case 1: $\exists i \geq 0, w_1, w_2 \in FORM_i$

$$w_1, w_2 \in FORM_i \to (w_1 \vee w_2) \in FORM_{i+1} \to (w_1 \vee w_2) \in INDFORM$$

Case 2: $\nexists i \geq 0, w_1, w_2 \in FORM_i$. WLOG, suppose $w_1 \in FORM_i, w_2 \in FORM_j, i < j$

$$\forall w \in FORM_n, w \in FORM_{n+1} \text{By induction } w_1 \in FORM_i \to w_1, w_2 \in FORM_j$$

Back to case 1.

□

7.2

$$\forall \text{(P,C)-closed set } S, INDFORM \subseteq S$$

To prove this, we only need to to prove that $INDFORM \subseteq S_{min} = \bigcap_{S \text{ is (P,C)-closed}} S$.

Proof. Base case:

$$\bigcup_{i=0}^{0} FORM_i = FORM_0 = P \subseteq S_{min}$$

Inductive hypothesis: Suppose

$$\bigcup_{i=0}^{n} FROM_i \subseteq S_{min}$$

$$\begin{aligned}
&\bigcup_{i=0}^{n} FROM_i \subseteq S_{min} \to FORM_n \in S_{min} \\
&\quad S_{min} \text{ is (P,C)-closed} \to \forall i \in [1, n], \forall \phi \in FORM_n, (\neg \phi) \in S_{min}, \\
&\quad \forall i \in [0, n], \forall \phi_1, \phi_2 \in FORM_i, (\phi_1 \vee \phi_2) \in S_{min} \\
&\to \bigcup_{i=0}^{n} FROM_i \cup \{(\neg \phi) \mid \forall \phi \in \bigcup_{i=0}^{n} FROM_i\} \cup \{(\phi_1 \vee \phi_2) \mid \forall \phi_1, \phi_2 \in \bigcup_{i=0}^{n} FROM_i\} \subseteq S_{min} \\
&\to \bigcup_{i=0}^{n+1} FROM_i \subseteq S_{min}
\end{aligned}$$

□

Definition 4.16 (Alternative Inductive Definition)**.** *Define* $AltFORM_0 = P$ *and for* $i \geq 0$*:*

$$\begin{aligned} AltFORM_{i+1} = \{(\neg w) \mid w \in AltFORM_i\} \\ \cup \{(w_1 \vee w_2) \mid w_1 \in AltFORM_i,\ \exists j \leq i,\ w_2 \in AltFORM_j\} \\ \cup \{(w_1 \vee w_2) \mid w_2 \in AltFORM_i,\ \exists j \leq i,\ w_1 \in AltFORM_j\}. \end{aligned}$$

Define

$$AltINDFORM = \bigcup_{i \geq 0} AltFORM_i.$$

Question 8

1. Show that $FORM_i = \bigcup_{j=0}^{i} AltFORM_j$ for all i.
2. Show that $INDFORM = AltINDFORM$.

8.1

Lemma 4.17.

Proof. Base case:

$$FORM_0 = \bigcup_{j=0}^{0} AltFORM_j$$

$$FORM_0 = P = AltFORM_0$$

Inductive hypothesis: Suppose

$$\forall j \in [0, i], FORM_i = \bigcup_{k=0}^{i} AltFORM_k$$

$$\begin{aligned} FORM_{i+1} &= \bigcup_{k=0}^{i} AltFORM_k \cup \{(\neg\phi) \mid \phi \in \bigcup_{k=0}^{i} AltFORM_k\} \cup \{(\phi_1 \vee \phi_2) \mid \phi_1, \phi_2 \in \bigcup_{k=0}^{i} AltFORM_k\} \\ &\supseteq \bigcup_{k=0}^{i} AltFORM_k \cup \{(\neg\phi) \mid \phi \in AltFORM_i\} \cup \{(\phi_1 \vee \phi_2) \mid \phi_1, \phi_2 \in \bigcup_{k=0}^{i} AltFORM_k\} \end{aligned}$$

$$\bigcup_{k=0}^{i+1} AltFORM_k \subseteq FORM_{i+1}$$

Now we prove the other way:

$$\forall \phi \in FORM_{i+1}, \phi \in \bigcup_{k=0}^{i} AltFORM_k \text{ or } \phi = (\neg\psi), \psi \in \bigcup_{k=0}^{i} AltFORM_k \text{ or } \phi = (\psi_1 \vee \psi_2), \psi_1, \psi_2 \in \bigcup_{k=0}^{i} AltFORM_k$$

Case 1:

$$\phi \in \bigcup_{k=0}^{i} AltFORM_k \rightarrow \phi \in \bigcup_{k=0}^{i+1} AltFORM_k$$

Case 2:

$$\psi \in \bigcup_{k=0}^{i} AltFORM_k \rightarrow \exists j \leq i, \psi \in AltFORM_j$$
$$\phi = (\neg\psi), \psi \in AltFORM_j, \phi \in AltFORM_{j+1}$$
$$\rightarrow \phi \in \bigcup_{k=0}^{i+1} AltFORM_k$$

Case 3:

$$\psi_1, \psi_2 \in \bigcup_{k=0}^{i} AltFORM_k \rightarrow \exists j, l \leq i, \psi_1 \in AltFORM_j, \psi_2 \in AltFORM_l$$
$$\phi = (\psi_1 \vee \psi_2), \psi_1 \in AltFORM_j, \psi_2 \in AltFORM_l, \phi \in AltFORM_{j+1}$$
$$\rightarrow \phi \in \bigcup_{k=0}^{i+1} AltFORM_k$$

$$FORM_{i+1} \subseteq \bigcup_{k=0}^{i+1} AltFORM_k \rightarrow FORM_{i+1} = \bigcup_{k=0}^{i+1} AltFORM_k$$

□

Formation Sequences

Definition 4.18 (Formation Sequence)**.** *A (P, C)-formation sequence is a nonempty finite sequence $\rho = w_1, \ldots, w_k$ such that for each i:*

- *$w_i \in P$, or*
- *$w_i = (\neg w_j)$ for some $j < i$, or*
- *$w_i = (w_j \vee w_k)$ for some $j, k < i$.*

The sequence belongs to w if $w_k = w$.

Q9

Give two distinct formation sequences that belong to

$$w = ((\neg p_1) \vee ((\neg p_3) \vee (\neg(p_2 \vee (\neg p_1)))))$$

Two distinct formation sequences ρ_1, ρ_2 are:

ρ_1**:**

$$
\begin{aligned}
w_1 &= p_1 \\
w_2 &= (\neg w_1) \\
w_3 &= p_2 \\
w_4 &= (w_3 \vee w_2) \\
w_5 &= (\neg w_4) \\
w_6 &= (p_3) \\
w_7 &= (\neg w_6) \\
w_8 &= (w_7 \vee w_5) \\
w_9 &= (w_2 \vee w_8)
\end{aligned}
$$

ρ_2**:**

$$
\begin{aligned}
w_1 &= p_1 \\
w_2 &= p_2 \\
w_3 &= p_3 \\
w_4 &= (\neg w_3) \\
w_5 &= (\neg w_1) \\
w_6 &= (w_2 \vee w_5) \\
w_7 &= (\neg w_6) \\
w_8 &= (w_4 \vee w_7) \\
w_9 &= (w_5 \vee w_8)
\end{aligned}
$$

These are distinct because the components at index $i = 2$ differ ($(\neg p_1)$ vs p_2).

Lemma 4.19. *$INDFORM = \{w \in \Sigma^* \mid w$ is formation-sequence-well-formed$\}$.*

Q10

Prove the lemma by showing:

1. Every formation-sequence-well-formed string is in $INDFORM$.
2. Every $w \in INDFORM$ has a formation sequence.

Before attemping Q10 from the above definition of formation-sequnce-well-formed, I will prove the following lemma using the definition below first:

Definition 4.20. *Define $ExFORM_0 = P$*

$$
\begin{aligned}
ExFORM_{i+1} = (&\{(\neg w) \mid w \in ExFORM_i\} \\
&\cup \{(w_1 \vee w_2) \mid w_1 \in ExFORM_i, \exists j \leq i, w_2 \in ExFORM_j\} \\
&\cup \{(w_1 \vee w_2) \mid w_2 \in ExFORM_i, \exists j \leq i, w_1 \in ExFORM_j\} \\
&) \setminus \bigcup_{j=0}^{i} ExFORM_j
\end{aligned}
$$

Lemma 4.21.

$$\bigcup_{j=0}^{i} ExFORM_j = FORM_i$$

Proof. Base case:

$$ExFORM_0 = P = FORM_0$$

Inductive Hypothesis: Suppose $\forall n \geq 0$,

$$\bigcup_{i=0}^{n} ExFORM_i = FORM_n$$

From definition 2.19,

$$\begin{aligned}
FORM_{n+1} = \bigcup_{i=0}^{n} ExFORM_i \cup \{(\neg w) \mid w \in \bigcup_{i=0}^{n} ExFORM_i\} \\
\cup \{(w_1 \vee w_2) \mid w_1 \in \bigcup_{i=0}^{n} ExFORM_i, \exists j \leq i, w_2 \in \bigcup_{i=0}^{n} ExFORM_j\} \\
\cup \{(w_1 \vee w_2) \mid w_2 \in \bigcup_{i=0}^{n} ExFORM_i, \exists j \leq i, w_1 \in \bigcup_{i=0}^{n} ExFORM_j\}
\end{aligned}$$

Definition 4.22.

$$\begin{aligned}
FromFORM : n \mapsto \{(\neg w) \mid w \in \bigcup_{i=0}^{n} ExFORM_i\} \\
\cup \{(w_1 \vee w_2) \mid w_1 \in \bigcup_{i=0}^{n} ExFORM_i, \exists j \leq i, w_2 \in \bigcup_{i=0}^{n} ExFORM_j\} \\
\cup \{(w_1 \vee w_2) \mid w_2 \in \bigcup_{i=0}^{n} ExFORM_i, \exists j \leq i, w_1 \in \bigcup_{i=0}^{n} ExFORM_j\}
\end{aligned}$$

Lemma 4.23.

$$FORM_{n+1} = \bigcup_{i=0}^{n} ExFORM_i \cup FromFORM(n)$$

Lemma 4.24.

$$ExFORM_{n+1} = (\bigcup_{i=0}^{n} ExFORM_i \cup FromFORM(n)) \setminus \bigcup_{i=0}^{n} ExFORM_i$$

$$ExFORM_{n+1} = FromFORM(n) \setminus \bigcup_{i=0}^{n} ExFORM_i$$

From Lemma 4.23 and Lemma 4.24, we have

$$\begin{aligned}
\bigcup_{i=0}^{n+1} ExFORM_i &= \bigcup_{i=0}^{n} ExFORM_i \cup ExFORM_{n+1} \\
&= \bigcup_{i=0}^{n} ExFORM_i \cup (FromFORM(n) \setminus \bigcup_{i=0}^{n} ExFORM_i) \\
&= \bigcup_{i=0}^{n} ExFORM_i \cup FromFORM(n) \\
&= FORM_{n+1}
\end{aligned}$$

$\square$

Lemma 4.25.
$$\bigcup_{i\geq 0} ExFORM_i = \bigcup_{i\geq 0} FORM_i$$

Proof. $\bigcup_{i\geq 0} ExFORM_i \subseteq \bigcup_{i\geq 0} FORM_i$

$\forall n \geq 0$, From Lemma 2.28

$$\phi \in \bigcup_{i=0}^{n} ExFORM_i \to \phi \in FORM_n \to \phi \in \bigcup_{i=0}^{n} FORM_i$$

$\bigcup_{i\geq 0} ExFORM_i \supseteq \bigcup_{i\geq 0} FORM_i$

Corollary from Lemma 2.28 $\square$

Corollary 4.26.
$$\bigcup_{i\geq 0} ExFORM_i = INDFORM$$
$$\forall \phi \in ExFORM_i \to \phi \textit{ is well-formed formula}$$

10.1, 2

Claim:

$$FORMSeq = \{w \in \Sigma^* \mid w \text{ is formation-sequence-well-formed}\} = INDFORM$$

Proof. Let ρ_i denotes the set of ρ that contains a sequence of i strings $w_1, w_2, \ldots, w_i$
Define

$$str : \rho \mapsto w, str(\rho = w_1, w_2, \ldots, w_k) = w_k$$
$$str : \rho_i \mapsto \{w\}, str(\rho = w_1, w_2, \ldots, w_k) = \{w \mid w = str(\rho), \rho \in \rho_i\}$$

Claim:
$$\bigcup_{k\geq 1} str(\rho_k) = \bigcup_{i\geq 0} ExFORM_i$$

Proof. $\bigcup_{k\geq 1} \rho_k \subseteq \bigcup_{i\geq 0} ExFORM_i$

$\forall w \in FORMSeq, \exists \rho = w_1, w_2, \ldots, w_k, w = w_k$
Base case:

$$\bigcup_{k=1}^{1} str(\rho_1) = \{w \mid w \in P\} = \bigcup_{i=0}^{0} ExFORM_i$$

Inductive hypothesis: Suppose

$$\bigcup_{j=1}^{k+1} str(\rho_j) \subseteq \bigcup_{i=0}^{k} ExFORM_i$$

$$\bigcup_{j=1}^{k+2} str(\rho_j) = \bigcup_{j=1}^{k+1} str(\rho_j) \cup \{w \mid \exists \rho \in \rho_{k+2}, w = str(\rho)\}$$

By definition, $w_{k+2} = str(\rho)$ for some $\rho \in \rho_{k+2}$, one of the 3 below holds:

1. $w_{k+2} \in P \to w_{k+2} \in \bigcup_{i \geq 0} ExFORM_i$

2. $w_{k+2} = (\neg w_n), n < k+2 \to w_n \in \bigcup_{j=1}^{n} str(\rho_j) \subseteq \bigcup_{i=0}^{k} ExFORM_i$

3. $w_{k+2} = (w_m \vee w_n), m, n < k+2 \to w_m, w_n \in \bigcup_{j=1}^{\max(m,n)} str(\rho_j) \subseteq \bigcup_{i=0}^{k} ExFORM_i$

$\bigcup_{k \geq 1} \rho_k \supseteq \bigcup_{i \geq 0} ExFORM_i$

Base case is the same as above.
Inductive hypothesis: (this will be used in Q11)

Lemma 4.27.

$$\bigcup_{i=0}^{k} ExFORM_i \subseteq \bigcup_{j=1}^{2(k+1)} str(\rho_j)$$

$\forall w \in ExFORM_{k+1}$, one of the 3 below holds: 1.

$$w \in P \to w \in ExFORM_0 \to w \in \bigcup_{i=0}^{k} ExFORM_i \subseteq \bigcup_{j=1}^{k+1} str(\rho_j)$$

2.

$$\begin{aligned} w = (\neg w_n), w_n \in ExFORM_k \to \exists \rho = w_1, w_2, \ldots, w_k \in \bigcup_{j=1}^{k+1} \rho_j, w_n = str(\rho) \\ \to \exists \rho' \in \rho_{k+2}, \rho' = w_1, w_2, \ldots, w_k, (\neg w_n) \\ \to w \in str(\rho_{k+2}) \to w \in \bigcup_{j=1}^{k+2} str(\rho_j) \\ \to w \in \bigcup_{j=1}^{2(k+1)} str(\rho_j) \end{aligned}$$

3. (Concat finite sequences that yield formula composing disjunctions gives finite sequence)

$$\begin{aligned} & w = (w_m \vee w_n), WLOG, w_m \in ExFORM_k, \exists q \leq k, w_n \in ExFORM_q \\ & \quad \to \exists \rho^m \in \rho_{k+1} = w_1^m, w_2^m, \ldots, w_k^m, \\ & \quad\quad \rho^n \in \rho_{q+1} = w_1^n, w_2^n, \ldots, w_q^n \in \bigcup_{j=1}^{k+1} \rho_j, w_m = str(\rho^m), w_n = str(\rho^n) \\ & \quad \to \exists \rho' = w_1^m, w_2^m, \ldots, w_k^m, w_1^n, w_2^n, \ldots, w_q^n, (w_k^m \vee w_q^n) \in \rho_{k+q+2} \\ & \quad \to w \in \bigcup_{j=1}^{k+q+1} str(\rho_j) \\ & \quad \to w \in \bigcup_{j=1}^{2(k+1)} str(\rho_j) \end{aligned}$$

Therefore,

$$\bigcup_{k \geq 1} str(\rho_k) = \bigcup_{i \geq 0} ExFORM_i$$

$\square$

By Corollary 4.26,

$$\bigcup_{k\geq 1} str(\rho_k) = INDFORM$$

□

Q11

Lemma 4.28. *Every well-formed formula ϕ is of one of the following forms:*

1. $\phi \in P$,
2. $\phi = (\neg\psi)$,
3. $\phi = (\psi_1 \vee \psi_2)$.

Proof. From Corollary 4.26,

$$\forall\phi \in INDFORM, \exists k \geq 0, \phi \in ExFORM_k$$

Case $x = 0$:

$$x = 0 \to \phi \in ExFORM_0 = P$$

Case $x > 0$:

$$\begin{aligned}
&\phi \in ExFORM_x \text{ and } \forall y \in [0,x), \phi \notin ExFORM_y \\
&\forall\phi \in ExFORM_x, \phi \in P \text{ or} \\
&\phi \in (\{(\neg\psi) \mid \psi \in ExFORM_{x-1}\} \\
&\qquad \cup \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_{x-1}, \exists j \leq x-1, \psi_2 \in ExFORM_j\})
\end{aligned}$$

Since we define $ExFORM_i$ to be pair wise disjoint, we guarantee that

$$\phi \in ExFORM_k \iff k \text{ is the minimum depth needed to form } \phi \text{ in the syntax tree}$$

Intuitively, x is the minimum depth required to formulate ϕ, thus with depth $d \in [0,x)$, ϕ cannot be formed
Suppose $\phi \in P$,

$$\phi \in ExFORM_0, \phi \in ExFORM_x, x > 0$$

Contradiction. Therefore

$$\phi \in (\{(\neg\psi) \mid \psi \in ExFORM_{x-1}\} \cup \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_{x-1}, \exists j \leq x-1, \psi_2 \in ExFORM_j\})$$

Now we will prove $\forall\phi \in INDFORM$

$$(\phi \in \{(\neg\psi) \mid \psi \in ExFORM_{x-1}\}) \oplus (\phi \in \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_{x-1}, \exists j \leq x-1, \psi_2 \in ExFORM_j\})$$

Suppose we enforce the use of parentheses for all subformula, that is except $\phi \in P$, all formula starts with $'('$. If for some i, $\phi \in \{(\neg\psi) \mid \psi \in ExFORM_i\}$, $'\neg'$ must be at index 1. Likewise, if for some $i, \phi \in \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_i, \exists j \leq i, \psi_2 \in ExFORM_j\}$, $'\neg'$ must not be at index 1. Thus $\nexists\phi \in INDFORM, \phi \in \{(\neg\psi) \mid \psi \in ExFORM_{x-1}\}$ and $\phi \in \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_i, \exists j \leq i, \psi_2 \in ExFORM_j\}$ □

Lemma 4.29 (Characterization of Well-Formed Formulae)**.** *Let P be a set of propositions and $C = \{\neg, \vee\}$. Every well-formed formula ϕ over (P, C) is of exactly one of the following forms:*

1. *$\phi \in P$,*
2. *$\phi = (\neg\psi)$ for some well-formed formula ψ,*
3. *$\phi = (\psi_1 \vee \psi_2)$ for some well-formed formulae ψ_1, ψ_2.*

Structural Induction Principle. A property $P(\phi)$ holds for all well-formed formulae if:

- $P(p)$ holds for all $p \in P$,
- $P(\psi)$ implies $P(\neg\psi)$,
- $P(\psi_1)$ and $P(\psi_2)$ imply $P(\psi_1 \vee \psi_2)$.

Q1

1.1

Theorem 4.30. *Let α be a well-formed formula. Let $c(\alpha)$ denote the number of occurrences of binary connective symbols $(\wedge, \vee, \rightarrow, \leftrightarrow)$ in α, and let $s(\alpha)$ denote the number of occurrences of sentence symbols in α. Then*

$$s(\alpha) = c(\alpha) + 1.$$

Proof. **Base Case:** Let $\alpha = p$ for some proposition $p \in \mathcal{P}$.

- $s(p) = 1$ and $c(p) = 0$.
- Since $1 = 0 + 1$, $P(p)$ holds for all atomic propositions.

Inductive Case 1 (Negation): Assume $P(\psi)$ holds, i.e., $s(\psi) = c(\psi) + 1$. Let $\alpha = (\neg\psi)$.

- $s(\neg\psi) = s(\psi)$ because no new sentence symbols are added.
- $c(\neg\psi) = c(\psi)$ because $\neg$ is a unary, not binary, connective.
- Thus, $s(\alpha) = s(\psi) = c(\psi) + 1 = c(\alpha) + 1$. $P(\alpha)$ holds.

Inductive Case 2 (Binary Connectives): Assume $P(\psi_1)$ and $P(\psi_2)$ hold. Let $\alpha = (\psi_1 c \psi_2)$ where $c \in \{\vee, \rightarrow\}$.

- $s(\alpha) = s(\psi_1) + s(\psi_2)$ (sum of occurrences in sub-formulae).
- $c(\alpha) = c(\psi_1) + c(\psi_2) + 1$ (sum of binary connectives plus the new $\star$).
- By the inductive hypothesis: $s(\alpha) = (c(\psi_1) + 1) + (c(\psi_2) + 1)$.
- Rearranging gives $s(\alpha) = (c(\psi_1) + c(\psi_2) + 1) + 1$.
- Therefore, $s(\alpha) = c(\alpha) + 1$. $P(\alpha)$ holds.

By structural induction, the property holds for all well-formed formulae. □

1.2

With only binary connectors, we show by induction: Base: $\forall p \in P, |p| = 1$
Inductive hypothesis:

Q2

Recall the following derived connectives:

- $\phi_1 \wedge \phi_2 \equiv \neg\big((\neg\phi_1) \vee (\neg\phi_2)\big)$,
- $\bot \equiv \neg(p \vee \neg p)$ for some $p \in P$,
- $\phi_1 \to \phi_2 \equiv (\neg\phi_1) \vee \phi_2$.

Using structural induction, show that for every formula ϕ over $(P, \{\neg, \vee\})$ with $P \neq \emptyset$, there exists a formula ψ using only connectives $\{\bot, \to\}$ such that ϕ and ψ are logically equivalent.

Proof. We will show that every formula formed over $\{\neg, \vee\}$ can be formed using $\{\bot, \to\}$, and denote the logically equivalent formula using the later connector set from ϕ as ϕ' Base case: $\forall p \in P, p' = p$
Inductive hypothesis: Suppose $\square$

Q3

Definition 4.31 (Conjunctive Normal Form)**.** *A formula ϕ over $(P, \{\neg, \vee, \wedge\})$ is in* conjunctive normal form (CNF) *if*

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m \quad (m \geq 1),$$

where each clause C_i has the form

$$C_i = \ell_{i,1} \vee \ell_{i,2} \vee \cdots \vee \ell_{i,k_i},$$

and each literal $\ell_{i,j}$ is either p or $\neg p$ for some $p \in P$.

Using structural induction, show that for every well-formed formula ϕ, there exists a logically equivalent formula ψ in CNF.

Q4

Definition 4.32 (Occurs)**.** *Define* occurs : FORM $\to \mathcal{P}(P)$ *inductively by:*

[leftmargin=2em]

- *If $\phi = p \in P$, then* $\text{occurs}(\phi) = \{p\}$,
- *If $\phi = \neg\psi$, then* $\text{occurs}(\phi) = \text{occurs}(\psi)$,
- *If $\phi = \psi_1 \oplus \psi_2$ with $\oplus \in \{\wedge, \vee\}$, then* $\text{occurs}(\phi) = \text{occurs}(\psi_1) \cup \text{occurs}(\psi_2)$.

Lemma 4.33 (Relevance Lemma)**.** *Let ϕ be a propositional formula and let $v_1, v_2 : P \to \{\text{true}, \text{false}\}$ be valuations such that $v_1(p) = v_2(p)$ for all $p \in \text{occurs}(\phi)$. Then $v_1 \models \phi$ if and only if $v_2 \models \phi$.*

Prove this lemma using structural induction.

Q5

Lemma 4.34 (Generalized Relevance Lemma)**.** *Let ϕ be a propositional formula and let v_1, v_2 be valuations such that $v_1(p) = v_2(p)$ for all p in some set $S \supseteq \text{occurs}(\phi)$. Then $v_1 \models \phi$ if and only if $v_2 \models \phi$.*

Prove this lemma.

Q6

Lemma 4.35 (Relevance Lemma for Logical Equivalence)**.** *Let ϕ_1, ϕ_2 be propositional formulae. The following are equivalent:*

[leftmargin=2em]

- *For all valuations v_1, v_2 agreeing on $\text{occurs}(\phi_1) \cup \text{occurs}(\phi_2)$, we have $v_1 \models \phi_1$ iff $v_2 \models \phi_2$.*
- *ϕ_1 and ϕ_2 are logically equivalent.*

Prove this lemma.

Q7

Let $G = (V, E)$ be an undirected graph and $k \in \mathbb{N}$. Construct a propositional formula ϕ_G such that:

1. $|\phi|$ is polynomial in $k + |V| + |E|$,
2. ϕ is satisfiable if and only if G has a vertex cover of size at most k.

Formally prove correctness of the encoding.

Proof. Define the proposition set P :

$$
\begin{aligned}
&P = P_{in} \cup P_{out} \cup P_{count} \\
&P_{in} = \{p_{v,in} \mid v \in V\} \\
&P_{out} = \{p_{v,out} \mid v \in V\} \\
&P_{count} = \{c_{i,j} \mid i, j \in [0, |V|]\} \\
&|P| \in O(|V|^2)
\end{aligned}
$$

There are constraint we need to consider:

$\forall v \in V, v \in V'$ **or** $v \in V \setminus V'$

$$\phi_{in} = \bigwedge_{v \in V} (p_{v,in} \vee p_{v,out})$$

$\forall v \in V, v \notin V'$ **or** $v \notin V \setminus V'$

$$\phi_{out} = \bigwedge_{v \in V} \neg(p_{v,in} \wedge p_{v,out})$$

All edges covered

$$\phi_e = \bigwedge_{\{u,v\} \in E} (p_{u,in} \vee (p_{v,in}))$$

$|V'| \leq k$

We use the following facts to define inductive rules:

- The number of true proposition is non decreasing as i increases in the sequence $p_1, \ldots, p_i$. $\rightarrow$ if the $c_{i,j}$ is true, $\forall \ell \geq i, c_{\ell,j}$ is true.
- if the $c_{i,j}$ is false, $\forall \ell \geq j, c_{i,\ell}$ is false

Inductively define $c_{i,j}$:

$$[c_{i-1,j} \vee (c_{i-1,j-1} \wedge p_{i,in})] \leftrightarrow c_{i,j}$$

We use bidirectional implication to prevent vacuous truthity of $c_{i,j}$ when conditions do not hold. And the last constraint can be represented by $\neg(c_{|V|,k+1})$
Putting everything together we can encode k-vertex cover using

$$\phi = \phi_{in} \wedge \phi_{out} \wedge \phi_e \wedge (\neg c_{|V|,k+1})$$

□

Correctness of encoding

To prove the correctness of the encoding, we will show:

Theorem 4.36. *ϕ is satisfiable $\rightarrow \exists k$-vertex cover on G*

Proof. ϕ is satisfiable $\rightarrow \exists v, v \models \phi, \rightarrow$

$$\begin{aligned} v &\models \phi_{in} \\ v &\models \phi_{out} \\ v &\models \phi_e \\ v &\models (\neg c_{|V|,k+1}) \end{aligned}$$

As per construction, this guarentees $\forall v \in V$, v is either in the vertex cover or exclusively; $\forall e = \{u, v\} \in E$, either u or v inclusively are in the vertex cover, and the size of the cover is at most k □

Theorem 4.37. *$\exists k$-vertex cover on $G \rightarrow \phi$ is satisfiable*

Proof. Suppose $G = (V, E)$ has vertex cover $V' \subseteq V, |V'| \leq k$, let $V'' = V \setminus V', \forall v \in V, v \in V$ □

Q8

Given a partial $N^2 \times N^2$ Sudoku grid S, construct a propositional formula ϕ_S of size polynomial in N^2 such that ϕ_S is satisfiable if and only if S has a completion to a valid Sudoku grid. Prove correctness.

Background: Compactness

Definition 4.38 (Satisfiable)**.** *A set Γ of propositional formulae is satisfiable if there exists a valuation v such that $v \models \Gamma$.*

Definition 4.39 (Finitely Satisfiable)**.** *A set Γ is finitely satisfiable if every finite $\Gamma' \subseteq \Gamma$ is satisfiable.*

Theorem 4.40 (Compactness Theorem)**.** *A set Γ of propositional formulae is satisfiable if and only if it is finitely satisfiable.*

Q11

Lemma 4.41. *If Γ is satisfiable, then Γ is finitely satisfiable.*

Proof. Γ is satisfiable $\implies \exists v, \forall \phi \in \Gamma, v \models \phi$
$\forall$ finite subset $\Gamma' \subseteq \Gamma, \forall \psi \in \Gamma', \psi \in \Gamma \implies \forall \Gamma' \subseteq \Gamma, \forall \psi \in \Gamma;, v \models \psi$ □

Q12

Definition 4.42 (P-complete)**.** Γ *is P-complete if for every $p \in P$, either $p \in \Gamma$ or $\neg p \in \Gamma$.*

Definition 4.43 (P-consistent)**.** Γ *is P-consistent if for no $p \in P$ do both p and $\neg p$ belong to Γ.*

Lemma 4.44. *If Γ is finitely satisfiable and P-complete, then Γ is P-consistent.*

1. Prove the lemma.

Proof. Suppose otherwise, Γ is P-complete, finitely satisfiable and NOT P-consistent, $\exists$finite set $\Gamma' \subseteq \Gamma, \exists p \in P, p \in \Gamma', (\neg p) \in \Gamma' \implies \Gamma'$ is not satisfiable, a contradiction. □

12.2

- $P = \{p, \}$
- $\Gamma = \emptyset$

12.3

- $P = \{p\}$
- $\Gamma = \{(\neg(p \vee (\neg p)))\}$

12.4

I cannot

Q13

Let Γ be finitely satisfiable and let ϕ be a formula.

1. Prove that at least one of $\Gamma \cup \{\phi\}$ or $\Gamma \cup \{\neg\phi\}$ is finitely satisfiable.
2. Give examples as requested.

Lemma 4.45. *at least one of $\Gamma \cup \{\phi\}$ or $\Gamma \cup \{\neg\phi\}$ is finitely satisfiable.*

Proof. We will prove the contrapositive of this statement: if neither $\Gamma \cup \{\phi\}$ nor $\Gamma \cup \{(\neg\phi)\}$ is satisfiable, Γ is not finitely satisfiable. Consider finite subset $\Gamma_1, \Gamma_2 \subseteq \Gamma$, and $\phi \in FORM, \Gamma_1 \cup \{\phi\}$ and $\Gamma_2 \cup \{(\neg\phi)\}$ are both not satisfiable, $\forall v, v \not\models (\Gamma_1 \cup \{\phi\}) \implies v \not\models \Gamma_1$ or $v \not\models \phi$, and $\forall v, v \not\models (\Gamma_2 \cup \{(\neg\phi)\}) \implies v \not\models \Gamma_2$ or $v \not\models (\neg\phi)$
Rewrite with respect to the set of all valuations, define

- $V_1 = \{v : P \mapsto \{T, F\} \mid v \models \Gamma_1\}$
- $V_2 = \{v : P \mapsto \{T, F\} \mid v \models \Gamma_2\}$
- $V_\phi = \{v : P \mapsto \{T, F\} \mid v \models \{\phi\}\}$

- $V_{(\neg\phi)} = \{v : P \mapsto \{T, F\} \mid v \not\models \{\phi\}\}$

$$V_1 \cap V_\phi = \emptyset \implies \forall v \in V_1, v \not\models \phi \implies v \models (\neg\phi) \implies V_1 \subseteq V_{(\neg\phi)}$$
$$V_2 \cap V_{(\neg\phi)} = \emptyset \implies \forall v \in V_2, v \not\models (\neg\phi) \implies v \models \phi \implies V_2 \subseteq V_\phi$$

Now consider also a finite subset to $\Gamma, \Gamma_1 \cup \Gamma_2$, define

- $V_\cup = \{v : P \mapsto \{T, F\} \mid v \models (\Gamma_1 \cup \Gamma_2)\}$

$V_\cup = (V_1 \cap V_2) \subseteq (V_\phi \cup V_{(\neg\phi)}) = \emptyset \implies \nexists v, v \models (\Gamma_1 \cup \Gamma_2) \implies \Gamma$ is not finitely satisfiable. □

13.2

- $\Gamma = \{p\}, \phi = q$

13.3

- $\Gamma = \{p, (\neg q)\}, \phi = q$

Q14, 15

Let $P = \{p_1, p_2, \dots\}$ and define $\Gamma_0, \Gamma_1, \dots$

Lemma 4.46. $\Gamma_U = \bigcup_{i \geq 0} \Gamma_i$ *is finitely satisfiable.*

Lemma 4.47.
$$\forall i \geq 0, \Gamma_i \textit{ is fintiely satisfiable}$$

Proof. Base case: $i = 0, \Gamma_i = \Gamma_0 = \Gamma$ is finitely satisfiable by definition.
Inductive Hypothesis: Γ_n is finitely satisfiable. By Lemma 4.45, $\Gamma_n \cup \{p_n\}$ is finitely satisfiable or $\Gamma_n \cup \{(\neg p_n)\}$ is finitely satisfiable. □

Proof. Consider finite set $\Gamma' \subseteq \Gamma_U$, since Γ' is finite, and $\forall k \geq 0, \Gamma_k \subseteq \Gamma_{k+1}$ by definition, $\exists j \geq 0, \Gamma' \subseteq \Gamma_j = \bigcup_{i \geq 0}^{j} \Gamma_i$. By Lemma 4.41, Γ_j is satisfiable $\rightarrow \Gamma'$ is finitely satisfiable. □

Proof. From the construction, $\forall i \geq 0$ at least one of p_i and $(\neg p_i)$ is in $\bigcup_{i \geq 0} \Gamma_i$. Since $\bigcup_{i \geq 0} \Gamma_i$ is finitely satisfiable, $\forall i \geq 0, p_i$ and $(\neg p_i)$ cannot both be in $\bigcup_{i \geq 0} \Gamma_i$. Otherwise $\{p_i\} \cup \{(\neg p_1)\} \subseteq \bigcup_{i \geq 0} \Gamma_i$ is not satisfiable, contradiction. □

Now we have proven that the construction proposed from a finitely satisfiable formula set Γ to $\bigcup_{i \geq 0} \Gamma_i$ is also finitely satisfiable. Now to complete the proof of the compactness theorem, we need to prove that $\exists v, v \models \Gamma$.

Proof. From Lemma 4.46, $\forall i \geq 0, (p_i \in \Gamma_U) \oplus ((\neg p_i) \in \Gamma_U)$, $\exists v, v \models \Gamma_U$

$$v(p) : P \mapsto \{T, F\} = \begin{cases} T, & \text{if } p \in \Gamma_U \\ F, & \text{if } (\neg p) \in \Gamma_U \end{cases}$$

Define the literal set $Literals = \{p_i, (\neg p_i) \mid i \geq 0\}$, and function

$$\tau(p) : P \mapsto Literals = \begin{cases} p, & \text{if } p \in \Gamma_U \\ (\neg p), & \text{if } (\neg p) \in \Gamma_U \end{cases}$$

Lemma 4.48.
$$v \models \Gamma_U \rightarrow v \models \tau(\Gamma)$$

Let $\phi \in \Gamma$, Define literal set $X = \{\tau(p) \mid p \in OCCUR(\phi)\}$ where X is a set containing single proposition literals. By definition we have

$$X \subseteq \Gamma_U \to v \models X \to \forall \psi \in X, v(\psi) = T$$

and $OCCUR(X) = OCCUR(\phi)$ Since both X is finite, $X \cup \{\phi\}$ is a finite subset to Γ_U, thus satisfiable. $\to \exists v', v' \models X \cup \{\phi\} \to v' \models X, v' \models \phi \to \forall \psi \in X, v'(\psi) = T$.

$$\forall p \in OCCUR(\phi), v(p) = v'(p), v' \models \phi \to v \models \phi$$

$\square$

Q16

Solve Questions 10 and 11 on page 53 (end of Section 1.5) in *A Mathematical Introduction to Logic* by Herbert B. Enderton.